WORKED SOLUTION

STEP II 2006, Question 10

Pure • Collision

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q10

The key idea

Each collision is a two-equation problem: conservation of momentum along the line, and Newton's law of restitution (speed of separation equals e times speed of approach). Solve the A–B collision, then feed B's new velocity into the B–C collision. A second A–B collision happens exactly when A, which is behind B, ends up moving faster in the forward direction than B does after B has hit C. Part (ii) is then a pursuit problem with constant velocities.

Solution

We model A, B and C as particles moving along a straight line on a smooth surface, so no horizontal forces act except during collisions. Take the direction in which A is projected as positive. All velocities below are signed.

Part (i)

Collision of A and B. Let A and B have velocities v_A and v_B just afterwards. Momentum and restitution (with $e = \frac{1}{2}$) give

$$mv_A + km\,v_B = mu, \qquad v_B - v_A = \tfrac{1}{2}u.$$

Substituting $v_A = v_B - \frac{1}{2}u$ into the first equation gives $(1+k)v_B = \frac{3}{2}u$, so

$$v_B = \frac{3u}{2(1+k)}, \qquad v_A = v_B - \frac{u}{2} = \frac{(2-k)\,u}{2(1+k)}.$$

Since $v_B > 0$, B does move on towards C, and since $v_A < v_B$, A cannot catch B before B reaches C.

Collision of B and C. Let B and C have velocities w_B and w_C just afterwards. With $e = \frac{1}{4}$,

$$km\,w_B + 3m\,w_C = km\,v_B, \qquad w_C - w_B = \tfrac{1}{4}v_B.$$

Substituting $w_C = w_B + \frac{1}{4}v_B$ gives $(k+3)w_B = (k-\frac{3}{4})v_B$, so

$$w_B = \frac{(4k-3)\,v_B}{4(k+3)} = \frac{3(4k-3)\,u}{8(1+k)(k+3)}.$$

Also $w_C > w_B$, so B and C separate and never meet again. After this collision, nothing else acts on A or B until they meet, if they ever do.

? **Ask yourself** What exactly is the condition for A and B to collide again?

Once B has hit C, both A and B move with constant velocities, and A is behind B (it is on the negative side of B). The gap between them shrinks exactly when $v_A > w_B$, and then it shrinks to zero in finite time. If $v_A \le w_B$ the gap never decreases. This one inequality covers every sign case: for instance, if A is moving backwards but B is moving backwards even faster, B still catches A, and that is also $v_A > w_B$.

So we need

$$\frac{(2-k)u}{2(1+k)} > \frac{3(4k-3)u}{8(1+k)(k+3)}.$$

Multiplying by $8(1+k)(k+3)/u$, which is positive since $k > 0$, gives the equivalent inequality

$$\begin{aligned} 4(2-k)(k+3) &> 3(4k-3) \\ 24 - 4k - 4k^2 &> 12k - 9 \\ 4k^2 + 16k - 33 &< 0 \\ (2k-3)(2k+11) &< 0. \end{aligned}$$

Since $2k+11 > 0$, this holds if and only if $2k-3 < 0$. Because k is a mass ratio, $k > 0$. Hence A and B collide for a second time exactly when

$$0 < k < \tfrac{3}{2}.$$

At $k = \frac{3}{2}$ we get $v_A = w_B = \frac{u}{10}$: the two move on together at the same speed and never actually meet again, so the endpoint is excluded.

Part (ii)

With $k = 1$ the formulas above give

$$v_A = \frac{u}{4}, \qquad v_B = \frac{3u}{4}, \qquad w_B = \frac{3u}{64}, \qquad w_C = \frac{15u}{64}.$$

Put the origin where A and B first collide (that is, at B's starting point), and measure time t from that collision.

Up to the B–C collision. B covers the distance d to C at speed $\frac{3}{4}u$, which takes

$$t_1 = \frac{d}{3u/4} = \frac{4d}{3u}.$$

In this time A has moved $\frac{u}{4} \cdot \frac{4d}{3u} = \frac{d}{3}$. So A is at $\frac{d}{3}$ and B is at d: the gap is $\frac{2d}{3}$.

The chase. Now A moves at $\frac{u}{4} = \frac{16u}{64}$ and B at $\frac{3u}{64}$, so A closes the gap at the relative speed $\frac{13u}{64}$. That takes

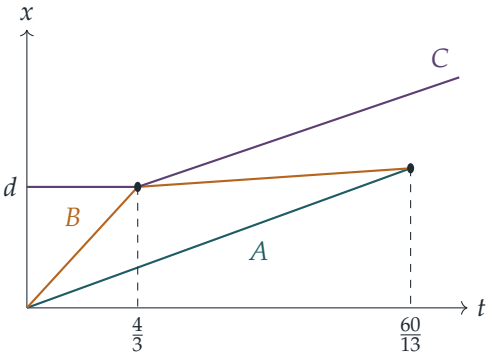
$$t_2 = \frac{2d/3}{13u/64} = \frac{128d}{39u}.$$

Total. The time between the two A–B collisions is

$$t_1 + t_2 = \frac{4d}{3u} + \frac{128d}{39u} = \frac{52d + 128d}{39u} = \frac{180d}{39u} = \frac{60d}{13u},$$

as required.

The distance–time graph below shows the motion, with $u = d = 1$. The lines for A and B meet again at $t = \frac{60}{13}$.



Checks

- Momentum: with $k = 1$, after the first collision $m \cdot \frac{u}{4} + m \cdot \frac{3u}{4} = mu$, and after the second $m \cdot \frac{3u}{64} + 3m \cdot \frac{15u}{64} = \frac{48}{64}mu = m \cdot \frac{3u}{4}$. Both balance.
- The boundary $k = \frac{3}{2}$: both v_A and w_B come out as $\frac{u}{10}$, which confirms the critical value.
- Sanity check: for large k, A bounces backwards ($v_A < 0$) while B continues forwards, so there's clearly no second collision. This agrees with the range.
- We checked numerically that $v_A > w_B$ exactly when $k < \frac{3}{2}$ for 10^5 random rational k, and a time-stepped simulation with $u = d = 1$ gave the second collision at $t \approx 4.6154$, matching $\frac{60}{13} \approx 4.6154$.

Common slips

- Using speeds instead of signed velocities in the restitution law. With signed velocities, separation speed is "front minus back", and the sign of v_A takes care of itself.

- Requiring $v_A > 0$ as part of the condition. A could in principle move backwards and still be caught by B (if $w_B < v_A$). The single condition $v_A > w_B$ is the right one.
- In (ii), timing the chase from the first collision instead of from the moment B's velocity changes, so missing the $\frac{d}{3}$ head start that A has already covered.

Beyond STEP

Momentum and restitution together make every collision a linear map from velocities before to velocities after. Chains of collisions, such as Newton's cradle or the famous problem of counting collisions between two blocks and a wall, are then studied by composing these maps. In first-year mechanics this leads to the centre-of-mass frame, where the restitution law becomes a simple scaling of relative velocity, and on to collisions in two and three dimensions. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and the centre-of-mass frame clearly.

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